

6 (**long**) The steel column shown in Fig.6 has a solid square cross section of side length 25 mm. The column, of length $L = 1$ m, is subjected to a compressive axial load P . The column is restrained from horizontal movement at its top, bottom and centre and is constrained to only move in the $x - y$ plane.

- (a) Using standard formulae, calculate the buckling load P_E of the column. [5]
- (b) If the column has an initial sinusoidal imperfection in the y direction given by $v_0(x) = \delta \sin \frac{2\pi x}{L}$, derive the differential equation for the shape of the column and hence calculate the total deflected shape of the beam. [20]
- (c) If the magnitude of the initial imperfection δ is 2 mm, calculate the maximum compressive stress in the column at a load $P = \frac{P_E}{2}$ and comment on its magnitude relative to the average compressive stress. [5]

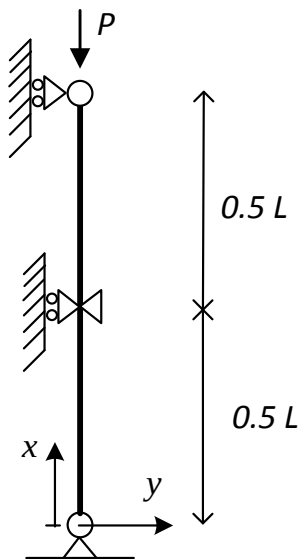


Fig. 6